

Fifty Years since the Coleman Report: Rethinking the Relationship between Schools and Inequality

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Abstract

In the half century since the 1966 Coleman Report, scholars have yet to develop a consensus regarding the relationship between schools and inequality. The Coleman Report suggested that schools play little role in generating achievement gaps, but social scientists have identified many ways in which schools provide better learning environments to advantaged children compared to disadvantaged children. As a result, a critical perspective that views schools as engines of inequality dominates contemporary sociology of education. However, an important body of empirical research challenges this critical view. To reconcile the field's main ideas with this new evidence, we propose a *refraction* framework, a perspective on schools and inequality guided by the assumption that schools may shape inequalities along different dimensions in different ways. From this more balanced perspective, schools might indeed reproduce or exacerbate some inequalities, but they also might compensate for others—socioeconomic disparities in cognitive skills in particular. We conclude by discussing how the mostly critical perspective on schools and inequality is costly to the field of sociology of education.

Keywords

stratification, schools, seasonal comparisons, compensation, reproduction

The authors of the 1966 Coleman Report famously concluded that variation in academic performance was strongly linked to children's family environments but hardly at all to per pupil expenditures or other measurable school characteristics (Coleman et al. 1966). In the decades following the report, many social science scholars resisted this position and articulated the ways in which schools reinforce or even promote inequality (e.g., Bourdieu 1977; Bowles and Gintis 1976; Cookson and Persell 1985; Kozol 1992). This "critical perspective" uncovered a wide range of school practices favoring the advantaged.

More recent scholarship has called the critical perspective into question. Compelling evidence from studies comparing school-year growth to summer growth suggests that socioeconomic achievement gaps in cognitive skills are more

a product of factors outside of schools than pernicious school processes (Downey, von Hippel, and Broh 2004; Entwisle and Alexander 1992; Heyns 1978). The evidence that schools temper the growth of socioeconomic inequality in academic skills presents a serious challenge to the critical perspective. Black/white learning gaps, however, *may* grow faster when school is in session versus out, complicating the picture regarding how schools matter (Condron 2009). These patterns

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have led to a state of theoretical vertigo in the sociology of education; it remains unclear where the pendulum currently rests. Do schools simply reproduce preexisting inequalities, do they magnify them, or do they help reduce them?

THE CRITICAL VIEW OF SCHOOLS

From the perspective of a body of scholarship much too expansive to review in detail here, schools either reproduce or exacerbate the inequalities that students bring with them (for useful overviews, see MacLeod 1995; Mehan 1992). Bowles and Gintis's (1976) argument that schools provide the capitalist economy with workers who know their place and are prepared for their roles, along with Bourdieu's (1977) perspective that schools favor the "cultural capital" of middle-class and elite students, form the basis of a "reproduction" paradigm, in which schools are seen as part of the process through which advantage and disadvantage are passed on from one generation to the next. Numerous sociological studies of education draw on this paradigm to investigate how schools reproduce inequalities (e.g., Paino and Renzulli 2013; Roscigno 1998).

Others go a step further and argue that schools do not simply reproduce inequality but rather exacerbate it. This position has gained stature for good reason, as scholars have documented a wide range of school practices and mechanisms that favor high socioeconomic status (SES) children. Prominent among them are "savage inequalities" in funding (Kozol 1992), which can lead to poorer school facilities, dated textbooks, and larger classrooms for disadvantaged children (Darling-Hammond 2010; Mosteller 1995). More recently, the narrative has shifted toward the unequal distribution of teacher quality in the wake of evidence from experimental studies suggesting that children improve 10 percentile points in a year if assigned a top- versus bottom-quartile teacher (Gordon, Kane, and Staiger 2006). In addition, residential segregation results in concentrations of disadvantaged children in the same schools, which has been shown to contribute to achievement gaps (Berends, Lucas, and Peñaloza 2008; Card and Rothstein 2007). Finally, curriculum differentiation practices within schools (e.g., ability grouping and tracking) might exacerbate preexisting skill differences as some children are exposed to more challenging material and more

effective learning environments than others (Condrón 2008; Gamoran 1992; Oakes 1985).

In our view, it is important to distinguish between whether schools reproduce inequality or exacerbate it. If schools reproduce inequality, then they pass on the inequalities received at kindergarten largely unchanged as children pass through the education system. Much of the "reproduction" literature, however, articulates an even more pernicious—exacerbatory—role for schools. Both sides describe a critical perspective on schools and inequality, but the two are distinct. And while reproductive and exacerbatory positions are sometimes mixed in the literature in a confusing way, a third possibility, that schools may *reduce* inequality, has been largely overlooked.

The Challenge to the Critical Perspective

The 1966 Equality of Educational Opportunity (EEO) study influenced a separate line of scholarship that placed less importance on schools themselves and greater emphasis on non-school sources of disparities in children's educational opportunities and outcomes. Often referred to as the Coleman Report, the study was commissioned by the Department of Education and involved more than 650,000 students and 4,000 schools. One finding that had the greatest influence on future work was summarized as follows (Coleman et al. 1966:325):

One implication stands out above all: That schools bring little influence to bear on a child's achievement that is independent of his background and general social context; and that this very lack of independent effect means that the inequalities imposed on children by their home, neighborhood, and peer environment are carried along to become the inequalities with which they confront adult life at the end of school.

Further supporting this claim, Jencks (1972:53) explored the same question with different data and agreed with the broad conclusion about schools: "Educational inequality does not explain cognitive inequality to any significant extent." Some misinterpreted Coleman's and Jencks's research as arguing that schools do not matter; but more

accurately, Coleman and Jencks posited that schools play only a minor role in shaping achievement gaps.

Even 10 years before Bowles and Gintis's (1976) book, Coleman and colleagues (1966) produced evidence suggesting non-school environments are the primary force behind achievement gaps. In subsequent decades, however, the sociology of education has focused more on school factors, and the Coleman Report fell out of favor, stemming in part from methodological criticisms. Critics pointed out, correctly, that Coleman and Jencks may have left unmeasured the most critical aspects of schools that generate inequality (e.g., teacher quality). Enthusiasm for understanding families' role in inequality was already diminished due to a hostile response to the 1965 Moynihan Report, which was critiqued as "blaming the victim" (Ryan 1976; see also Skrentny 2008). Ironically, while the Coleman Report initially was seen as conservative for implicating families and communities as the source of socioeconomic achievement gaps and Bowles and Gintis were seen as radically left, by the early 1980s, conservatives had embraced the view that schools are the problem (e.g., National Commission on Excellence in Education 1984).

Understanding How Schools Influence Inequality

It is difficult to adjudicate between the varying perspectives on schools and inequality without thinking carefully about the kind of evidence that is most useful. We identify three hurdles. First, children are influenced in important ways by both their families and their schools, so how do we separate the two? The average 18-year-old in the United States has spent just 13 percent of their waking hours in school (Walberg 1984), highlighting the importance of non-school environments. Children are not randomly distributed to schools, so it is difficult to know whether outcome differences between schools are a function of school processes or the widely disparate homes and neighborhoods where children spend most of their time. A common strategy scholars use to isolate school effects is to statistically equalize children across measurable dimensions of family background, such as socioeconomic status, family structure, and race/ethnicity. Another approach is to predict how much children change (e.g.,

learning gains over a year) rather than predict an outcome at one point in time. By predicting learning gains, scholars effectively start all children at zero and give schools no credit (or punishment) for children's beginning skills. These strategies are only partly effective, however, because why some children learn faster than others is largely unknown. For example, in models attempting to estimate as much variation in children's summer learning as possible, Burkam and colleagues (2004) used both methods and explained less than 15 percent of the variance, highlighting how most of the non-school characteristics that influence children's learning go unmeasured in large surveys. Studies attempting to understand how schools matter using these traditional techniques likely overestimate school effects because their statistical adjustments for non-school factors are insufficient.

The second hurdle for understanding how schools influence inequality is that we need to consider how all school processes—both exacerbatory and compensatory—stack up against each other. Some studies address the first problem (i.e., isolating school effects) by using random assignment, but these studies typically target a single school process (e.g., the Tennessee Project Star experiment on classroom size), which constitutes just one piece of the larger puzzle. This kind of research has value—we learn something about whether a particular school practice increases inequality—but it is limited for helping us understand schools' overall effect on inequality. We are left wondering whether the magnitude of all exacerbatory school processes outweighs that of compensatory ones (those that reduce inequality).

The third hurdle is crucial—even if we successfully isolate school effects and determine whether exacerbatory school processes outweigh compensatory ones, we still need to know whether *school* inequality outweighs *non-school* inequality. Scholars traditionally have framed the question narrowly, such as, "How well would a particular student perform if they attended school A versus school B?" This approach identifies variation in school quality, but it tells us little about how schools matter within the stratification system. Instead, we recommend that scholars broaden their scope by considering the counterfactual: "What would inequality look like if schools did not exist?" (Raudenbush and Eschmann 2015). This is important because schools may vary greatly in quality, but non-school environments

may vary even more (Downey et al. 2004). If this is the case, even unequal schools could be equalizing forces by reducing the level of inequality we would observe in their absence. The traditional counterfactual, however, does not account for that possibility.

Seasonal Comparisons

Seasonal comparison study designs offer one way of meeting these methodological hurdles. The seasonal nature of the U.S. school calendar—nine months of school followed by a three-month summer break—provides a natural experiment for understanding how schools matter (Gangl 2010). Similar to a crossover design in medical research, where patients are observed on and off treatment, seasonal comparison researchers observe how achievement gaps change when children are on treatment (in school) versus out (summer). This method circumvents the problem of trying to identify all the various school processes at stake. The overall consequence of all mechanisms (both exacerbatory and compensatory) is observable in how inequality changes when school is in session versus out of session. And seasonal comparisons provide a way of estimating what inequality would look like if children did not attend schools by leveraging the summer as an estimate of that counterfactual. Finally, unlike randomized experiments, which can be weak in generalizability, seasonal comparisons can be applied to nationally representative data, producing results strong in both internal and external validity.

Like all empirical methods, seasonal comparisons require assumptions. For example, the method assumes there is little treatment spillover. We must assume that school practices in the spring have little influence on summer learning so that the school treatment does not contaminate estimates of non-school learning. In practice, this assumption is nearly always violated because students are not tested precisely on the first and last days of school. Scholars try to reduce the extent of this problem by modeling the school segments of learning and subtracting them from estimates of summer learning. In addition, school practices in the spring, like sending children home with a summer reading list, could influence summer learning. Although there is little evidence of this kind of contamination among young children (Downey et al. 2004), it remains difficult to assess.

In addition, comparing whether group A, starting low on a scale, gains more than group B, starting high, requires interval-level scales where gains at the bottom are comparable to those at the top—like equal stair steps. Some of the better scales of cognitive skills may approach this requirement and reduce problems associated with ceiling effects that trouble longitudinal comparisons, but the field would benefit from greater discussion of this issue and perhaps greater use of nonparametric methods that do not depend on interval assumptions (Ho and Reardon 2011). Finally, are the patterns we observe in the summer really a good indicator of what would happen if there were no schooling? If parents knew their children would not return to school in the fall, would they behave the way they do in the summers? The field would benefit from a more energetic discussion of these assumptions. Nevertheless, we believe they are more plausible than the assumptions required for more traditional methods, that is, (1) models predicting learning gains and/or statistically controlling for observable differences in family background can successfully isolate school effects, (2) we know how all exacerbatory and compensatory mechanisms stack up against each other, and (3) school inequality is greater than non-school inequality. As a result, seasonal comparison studies prompt great interest.

What do we learn from seasonal comparison studies? Remarkably, they find that socioeconomic gaps in cognitive skills grow faster when school is out than when it is in. Evidence of this pattern began to emerge as early as the 1970s. Heyns's (1978) analysis of over 3,000 sixth and seventh graders in 42 Atlanta schools during the 1971 to 1972 school year and summer uncovered a groundbreaking key insight—children's learning during the summer is a product of non-school factors, whereas learning during the school year is a product of both school and non-school factors (for other early seasonal comparison studies, see Hayes and Grether 1983; Klibanoff and Haggart 1981; Murnane 1975). Observing how achievement gaps change between the seasons thus provides important leverage for understanding how schools matter. Subsequently, Entwisle and Alexander brought even greater attention to seasonal patterns with their analyses of children's progression through Baltimore schools in the 1980s (Alexander, Entwisle, and Olson 2007; Entwisle and Alexander 1992, 1994). The credibility of the seasonal pattern was bolstered by the first

nationally representative, seasonally collected data, the Early Childhood Longitudinal Study–Kindergarten Cohort of 1998 (ECLS-K) (Downey et al. 2004). The magnitude of the patterns differs from one study to another, but each has produced the same overall finding: Socioeconomic gaps in skills grow faster when school is out versus in. Alexander (1997:12) explains the ramifications of the seasonal comparison patterns: When it comes to inequality, “schooling is more ‘part of the solution’ than ‘part of the problem.’”

Scholars also have begun to apply seasonal methods to other dimensions of inequality, and the results are provocative. von Hippel and colleagues (2007) analyzed the ECLS-K data and found that children gained body mass index (BMI) roughly twice as fast in the summer than during the school year. Black, Hispanic, and low SES children were especially vulnerable to BMI gain during the summer months. Although it is tempting to infer that schools promote childhood obesity, given what is often available for children to eat and drink at school, clearly schools reduce the level of childhood obesity that we would observe in their absence. More recent analyses of the newer ECLS-K 2011 data replicate these patterns (von Hippel and Workman n.d.). Downey, Workman, and von Hippel (2016) analyzed the ECLS-K 2011 data for seasonal patterns of social and behavioral outcomes: Teachers rated children in terms of a set of learning behaviors (e.g., keeps belongings organized, shows eagerness to learn new things, works independently, easily adapts to changes in routine, persists in completing tasks, pays attention well, and follows classroom rules); gaps in these social and behavioral skills are large across SES, race, and gender at kindergarten entry but show little evidence of increasing faster when school is in versus out of session between kindergarten and second grade.

To date, the seasonal comparison method points to schools as compensatory when it comes to socioeconomic gaps in cognitive skills and children’s obesity and as neutral with respect to social/behavioral skills, but the patterns for race and cognitive skills are less clear. Heyns (1978) found that black/white gaps grew faster in the summer than in the school season in Atlanta, suggesting that schools are also compensatory with respect to race, but black/white patterns were less clear in Baltimore (Entwisle and Alexander 1994). And in the nationally representative ECLS-K data, the black/white gap grew faster

during kindergarten and first grade than during the summer in between, consistent with the view that schools generate inequality (Downey et al. 2004). Muddying the waters further, black/white skill disparities are well formed at kindergarten entry (Downey et al. 2004). Black/white gaps in cognitive skills appear to grow before kindergarten because of non-school factors, whereas the data from kindergarten through first grade implicate schools. There is also provocative evidence from the ECLS-K and Northwest Evaluation Association (NWEA) data that schools may undermine the performance of Asian Americans (Downey et al. 2004; Yoon and Merry 2015); the patterns for Hispanics are unclear. Overall, early seasonal studies suggested that schools are probably compensatory with respect to race, but the more recent analyses of larger-scale ECLS-K and NWEA data reveal more troubling patterns—white children may benefit more from school than do black or Asian children.

The evidence for the racial patterns leans in the direction of schools as exacerbatory, but the patterns for socioeconomic achievement gaps challenge the critical view of schools. Are schools driving some inequalities while simultaneously reducing others? Once seasonal comparison studies expand to consider outcomes beyond cognitive skills (e.g., health, graduation, and earnings), will schools’ pernicious effects become more apparent? We suggest that the time has come for a new framework for theorizing and analyzing schools and inequality that can accommodate the provocative compensatory seasonal patterns for socioeconomic gaps in cognitive skills and children’s obesity yet has the flexibility to consider how schools’ effect on inequality may vary across other dimensions of inequality.

THE REFRACTION FRAMEWORK

At the core of our framework is the idea that schools are “refractors” of inequality. Much like light is refracted when it enters a new medium (e.g., from air to water), we argue that inequalities are refracted when children enter schools. Light refracts in different ways, depending on whether it enters a slower, faster, or similar speed medium. Similarly, how inequality changes once children enter schools depends on how the new medium (schools) influences inequality’s trajectory vis-à-vis the previous medium (the non-school

environment). Schools' role could be (1) neutral (no change to inequality), (2) exacerbatory (makes inequality worse), or (3) compensatory (reduces inequality). One advantage of this perspective is that it elevates the compensatory possibility to a position where it can compete with the other, more dominant views. It also underscores how inequality is well established and on a meaningful trajectory prior to school entry, elevating the importance of early childhood experiences (Caudillo and Torche 2014; Heckman and Masterov 2007). We believe schools potentially can influence inequality in all three of these ways, but we focus here on the case for schools as compensatory because this view is the least developed.

Schools as Compensatory

In the nineteenth-century United States, education reformers argued that mass education would provide a common experience for children and therefore an avenue for social mobility (Giroux 2005). U.S. proponents of public schooling contrasted the system they were creating with their European counterparts, whose education system remained closely linked to social class. Horace Mann (1848), the well-known advocate for public schooling, wrote,

According to the European theory, men are divided into classes—some to toil and earn, others to seize and enjoy. According to the Massachusetts theory, all are to have an equal chance for earning, and equal security in the enjoyment of what they earn. . . . Education then, beyond all other devices of human origin, is a great equalizer of the conditions of men.

This view continues to have credibility among the general public, with some caveats. In the 1984 General Social Survey (GSS), the last year the question was asked, over 70 percent of respondents answered yes to the question: "Does everyone in this country have an opportunity to obtain an education corresponding to their abilities and talents?" The public view appears to be that schools serve as a vehicle for social mobility—the school system is a reasonably fair institution that rewards hard work. At the same time, however, most people acknowledge large disparities in school quality and try very hard to get their own children into

"good" schools (Johnson 2006). Although the public views the system as rewarding people fairly, they also put the onus of responsibility on individuals to find the best schools and work hard while in them.

In academia, seasonal comparison research provides the strongest evidence that schools are compensatory with regard to socioeconomic gaps in cognitive skills. Of course, it may not be clear why seasonal comparison evidence is viewed as supporting the compensatory position, given that high and low SES children typically learn at roughly the same rate when school is in session. Does that not suggest a neutral role for schools? In some cases, high SES children even learn a little faster than low SES children, yet scholars still conclude that schools are compensatory (e.g., Downey et al. 2004). If schools are not reducing achievement gaps in the absolute sense, and sometimes even allowing them to continue to grow a bit, how can schools be compensatory?

An analogy may help. Suppose we assess a year-long weight loss program by randomly assigning subjects to either treatment or control groups. And suppose that upon completion, the treatment group has not lost any weight. On its surface, this result suggests that the treatment failed. But the proper way to assess the causal effect of the treatment is in *comparison* to results for the control group. If the control group gained five pounds, on average, then the weight loss program had a positive effect even if the treatment subjects lost no weight. One can even imagine a scenario where the control group gained, on average, five pounds during the study while the treatment group gained two. Even though the treatment group experienced weight gain, we would still define the treatment as a success because it reduced the weight the treatment group would have gained in its absence. Similarly, the proper way to assess schools' effect on achievement gaps is not to focus solely on school year patterns but to *compare* the school year (treatment) and summer (control) patterns. Schools are compensatory whenever achievement trajectories are more equal when school is in versus out of session.

Compensatory Mechanisms

Given that many of the patterns from seasonal comparison research suggest that schools are compensatory with respect to socioeconomic gaps in

cognitive skills, we begin by considering possible school mechanisms that might reduce inequality. Our goal here is not to identify all compensatory school mechanisms. Rather, we merely initiate a discussion of school processes that are potentially compensatory when it comes to socioeconomic learning disparities.

First, schools may reduce achievement gaps via curriculum consolidation. Sociologists may be more familiar with the term *curriculum differentiation*, referring to school practices in which some children are exposed to different material and learning conditions than others, such as ability grouping, tracking, and retention practices. But schools also *consolidate* children's learning experiences, grouping them together even when their skills are disparate. For example, schools can organize children in many ways, but children's chronological age is the default basis upon which children are grouped. We do not disagree with that practice but note that the result is a powerful mechanism by which children of widely varying skills are exposed to the same curricular challenges. To understand how important this mechanism is for promoting equality, we can consider the distribution of cognitive skills among children in kindergarten, first, and fifth grades in the ECLS-K. We see substantial overlap in cognitive skills across grades: 40 percent of kindergartners outperform the bottom 10 percent of first graders in reading, and a nontrivial number of kindergartners read better than fifth graders (authors' calculations). What do schools tend to do with these high-performing kindergartners? A few are accelerated through the system by skipping a grade, but even this practice is unlikely to expose them to challenging material. Some of these children are so far ahead of their age-based peers that they would need to skip multiple grades to find curriculum in their sweet spot. Most remain with their age-group peers, a practice that is likely compensatory because it is difficult for these students to produce academic gains while exposed to material mostly below their level.

Second, schools may reduce achievement gaps by targeting resources toward disadvantaged children. It is important not to overlook the many education policies designed to improve school conditions for disadvantaged children. Title 1, Head Start, the Rehabilitation Act of 1973, and the Americans with Disabilities Act in 1990 all were intended to improve the quality of school experiences for low SES children (and to varying degrees have succeeded in doing so; see Ludwig

and Miller 2007). Most school funding comes from local and state sources, so a look at how resources are distributed across children with special needs versus honors students at the state level is informative. As one example, in 2007 in Kentucky, the average child with special needs received an additional \$11,970 per year, whereas the average child deemed gifted received an additional \$62 per year (Seiler et al. 2008).

Finally, it is possible that although much is made of the cultural mismatches between teachers and disadvantaged students, many teachers operate in a mostly egalitarian manner. The kinds of people attracted to teaching are distinct from the general population—they are more interested in helping others and more likely to endorse relatively egalitarian views. For example, in the GSS data, 47 percent of non-teachers said that “lack of effort” is a “very important” reason why some people are poor, compared to just 32 percent of teachers (authors' calculations). And a national survey of teachers found that when asked who was most likely to receive one-on-one attention, 80 percent of teachers said “academically struggling students” while just 5 percent said “academically advanced” students (Duffett, Farkas, and Loveless 2008). Teachers may favor the advantaged under some conditions, but overall, teacher behaviors exacerbating inequality may be outweighed by teachers' greater tendency to favor the disadvantaged.

THE COSTS OF IGNORING SCHOOLS' COMPENSATORY POTENTIAL

We believe that the best evidence currently indicates that schools play a meaningful compensatory role with respect to socioeconomic gaps in cognitive skills, and schools' compensatory role might extend further. If schools are playing a more significant compensatory role than is currently appreciated, what are the costs of this mistake?

The Cost to Research

Underestimating schools' compensatory role can lead researchers to overlook the important role that social conditions outside of school play in shaping the magnitude and malleability of achievement gaps. For example, it turns out that socioeconomic-based achievement gaps are

mostly formed prior to the onset of formal schooling. Perhaps the strongest evidence comes from the 1998 ECLS-K study, which provides the best data on a large sample of children followed from kindergarten through several years of school. The 90th to 10th income-based achievement gaps in reading grew about 12 percent between kindergarten and eighth grade, and the math gap actually narrowed (Reardon 2011; see also Duncan and Magnuson 2011). If socioeconomic-based achievement gaps form primarily prior to kindergarten and then increase only modestly (and sometimes even narrow) once children are in school, then most of the action regarding achievement gaps occurs in early childhood.

Another way to understand how broader social conditions matter is by comparing achievement gaps across countries. Significant cross-national work has explored variations in test scores for children already exposed to schools, but these studies mix the effects of school and non-school conditions in unknown ways. It is especially useful, therefore, to compare gaps in skills among children who have yet to start school. A study of U.S. and Canadian children's reading skills highlights this point. Merry (2013) documents how Canadians are ahead of U.S. children by a sizeable .30 standard deviation units (nearly a year's worth of learning) in the Programme for International Student Assessment (PISA) reading test given to 15- to 16-year-olds. Merry also compared similar-cohort Canadian and U.S. children on the Peabody Picture Vocabulary Test (PPVT) reading test at ages 4 to 5, before formal schooling had started, and found that a similar gap of .31 standard deviation units was already in place at that age.

This pattern prompts us to rethink school-based explanations for differences in international test scores and consider forces outside the control of the education system (e.g., access to health care, income inequality, racial/ethnic inequality, racial- and income-based housing segregation, the strength of organized labor, family structure, immigrant status, mass incarceration, the real value of the minimum wage, unemployment benefits, and family leave options). Merry's (2013) study raises the possibility that international differences in test scores among adolescents may have little (or maybe nothing) to do with schools. Perhaps Finland's impressive test scores, which have prompted great interest in their teachers and schools, are mostly a product of its successful social welfare programs.

Finally, it is unclear how conditions outside of schools shape schools' compensatory power. Downey and colleagues (2004) suggest that as inequality outside of school grows, schools' compensatory power also increases because the difference in school versus non-school environments becomes more acute—schools become an even more important haven for the disadvantaged. But does schools' compensatory power really increase, ad infinitum, as social conditions outside of schools become more disparate? One might expect a limit to this relationship—that when social conditions become highly unequal, children arrive at kindergarten with achievement gaps so large and entrenched that schools' compensatory power begins to wane. Have we reached that point? These kinds of contextual questions—how the gap at kindergarten entry changes over time, how it compares across countries, and how schools' compensatory power changes over time—are where sociologists shine, yet they have received insufficient attention.

The Cost to Policy

It is one thing if academics develop the wrong explanations for inequality; it is another if those explanations frame public discussions that lead to suboptimal policy advice. If we are currently underestimating schools' compensatory power to the extent that the seasonal comparison evidence suggests, then education policy is misinformed by underestimating the quality of schools serving low SES children. As a result, education scholars are likely overestimating the extent to which school reform is the most attractive mechanism for reducing achievement gaps. If socioeconomic gaps are mostly formed prior to formal schooling and change little after that, then it makes more sense to target early childhood policies that will prevent large gaps from emerging in the first place rather than focus on school policies aimed at remediating those gaps.

Yet, a significant portion of current education research emphasizes the importance of teacher quality (Goldhaber 2016; Hanushek 2010). We agree that some teachers raise children's skills more effectively than others, and improving teacher quality is a laudable goal. But what are the implications for achievement gaps? Most scholars assume that low SES children endure poorer teachers and that this plays an important

role in explaining gaps. If we measure teacher quality in terms of qualifications, experience, or whether teachers majored in the subject area they teach, then we find evidence consistent with this view—low SES children are exposed to poorer teachers than high SES children. But these observable teacher characteristics hardly predict students' learning. Most scholars now agree that when measuring teacher quality, we should focus on what really matters: how much students learn.

If we evaluate teacher or school quality via how much students learn, the evidence is more mixed regarding whether low SES children endure substantially poorer learning environments. Some data support the notion that high SES children enjoy more effective teachers. For example, a report from the Tennessee Department of Education, which uses a value-added approach for evaluating teachers, concludes that "Tennessee's teacher effectiveness data indicate that students in high poverty/high minority schools have less access to the 'most effective' teachers and more access to the 'least effective' teachers than students in low poverty/low minority schools" (Tennessee Department of Education 2007). A similar analysis in Louisiana found comparable patterns, although it did not replicate in Massachusetts (DeMonte and Hanna 2014).

The value-added models favored by many researchers, however, may not adequately isolate teachers' contributions to learning. One problem is that they usually gauge learning from one year to the next, for example, between the spring of third grade and the spring of fourth grade, and so the summer in between biases the estimate because high and low SES children's skills diverge for reasons unrelated to schools. Growth models constructed with 9-month data remove summer noise and correlate only around .50 with traditional growth models using 12-month data, demonstrating that summer noise is a nontrivial problem (Atteberry 2011).

There is not yet consensus over which kind of value-added model is most valid for isolating teacher or school effects, but as education scholars develop better techniques, one pattern stands out—models that more persuasively isolate these effects suggest that the differences in effectiveness between schools serving high and low SES children are modest or even nonexistent (Downey, von Hippel, and Hughes 2008; Lauen and Gaddis 2013). For example, analyzing a nationally representative sample of 287 schools, Downey and

colleagues found no relationship between their measure of school quality (i.e., the difference in schools' summer vs. school-year learning rates) and the percentage of children in the school receiving free or reduced lunches (Downey et al. 2008). Along the same lines, other work notes that private schools do not produce more learning than public ones (Lubienski and Lubienski 2013).

These studies lead to a remarkable conclusion—the distribution of teacher or school effectiveness (defined as promoting cognitive skills) may be only weakly related to socioeconomic status. The traditional story is that low SES children attend dramatically poorer schools and, as a result, are far behind their high SES peers. It is probably more accurate to say that low SES children arrive at kindergarten far behind but then mostly stop losing ground once in school. Current accountability schemes therefore most likely underestimate the performance of teachers and administrators serving disadvantaged children, leading to undeserved labels and sanctions. Parents and policy-makers also receive poor information about which schools are the best, undermining market mechanisms that might promote better schools (Downey et al. 2008).

If low SES children currently enjoy roughly similar learning environments relative to their high SES peers, what does this mean for reducing the achievement gap *via school reform*? It means that reducing gaps would require more than just raising the quality of schools serving low SES children to the level of those serving high SES children—it would require creating substantially *better* school learning environments for low SES children. This may be what happens when small-scale projects are able to close achievement gaps to some degree. For example, black children who were chosen via lottery for the Harlem Children's Zone and experienced several years in the program demonstrated eighth-grade skills on par with white children in New York City while their counterparts who were not chosen continued to lag behind (Dobbie and Fryer 2009). The Knowledge Is Power Program (KIPP) charter schools also have raised disadvantaged children's skills, in part by increasing the amount of time children spend in school and perhaps by attracting high-quality teachers ("Student Characteristics and Achievement" 2010). It is important to note that these occasional "high-flying" schools probably succeed by providing disadvantaged children with substantially *better* schools than their

advantaged peers attend. An obstacle for reducing inequality via school reform therefore is that doing so would require scaling up similar efforts and, importantly, making them available to low SES children while denying them to high SES children. Or, at the least, low SES children would need to enjoy greater benefits than high SES children from the school treatments provided.

Relatedly, one of the best investments U.S. policymakers could make would be to move away from the current, agriculturally based school calendar and extend the school year. Yet policymakers, influenced by popular school-based rhetoric, reasonably ask, "Why spend money increasing the quantity of schooling when the quality of schools serving disadvantaged children is so poor?" Scholars have promoted the view that schools serving low SES children are of very poor quality, thereby undermining support for increasing school exposure. This is unfortunate because a longer school year, with school quality distributed as it currently is, would improve the cognitive skills of U.S. children in general and benefit the disadvantaged the most (i.e., by lengthening the period of time in which they lose less ground to high SES children).

Of course, even if schools are already compensatory and not the source of socioeconomic gaps, they still may be an attractive policy lever for reducing gaps further. Through school reform, we can readily influence a broad range of children. But recall that these gaps emerge during early childhood and are nearly completely formed by kindergarten entry. When we consider school-based reforms for reducing achievement gaps, we need to know more about the cost-benefit analysis of broader societal policies that could prevent gaps from emerging in the first place versus school reforms aimed at remediating gaps. The question is not whether it is *possible* via school reform to reduce achievement gaps; the question is whether school reform is the *best strategy* for reducing them.

CONCLUSIONS

Fifty years after the Coleman Report, a critical view of schools has come to dominate the sociology of education, with scholars identifying a wide range of school mechanisms thought to reproduce or exacerbate inequality. The field is now at a crossroads. The dominant ideas continue to

support the notion that schools reproduce or exacerbate inequality, but compelling empirical evidence suggests that schools are compensatory when it comes to a key dimension of educational inequality—socioeconomic achievement gaps.

It may turn out that socioeconomic gaps in cognitive skills (and children's body mass index) are the exceptions, the only dimensions of inequality for which schools are compensatory. Maybe schools are an engine of socioeconomic inequality for most other outcomes. But if future work finds that schools promote inequality for most other outcomes, we would need to explain why the school mechanisms that reduce SES-based cognitive skill gaps do not extend to these other outcomes, highlighting the need to understand schools' compensatory mechanisms better. This gap in the sociology of education literature remains a significant obstacle.

Another possibility, and one we find more persuasive, is that schools are considerably more compensatory than previously thought. Coleman's conclusion—that schools play a largely neutral role in influencing achievement gaps—was wrong, but for a different reason than most critiques of his work suggest. Coleman may not have measured what matters about schools well, but he also did not measure the non-school environment well, and that error was even greater.

The primary need for change, in our view, lies in how education scholars frame their questions. Scholars tend to ask, are there school processes that favor advantaged children? If they find them, then they conclude that schools are culprits. We do not dispute that some school practices likely make inequality worse, and indeed we both have researched some of these practices ourselves (Condrón 2008; Downey and Pribesh 2004). But we contend that the literature typically fails to compare the *magnitude* of exacerbatory school mechanisms against that of compensatory ones, resulting in a one-sided view of schools and inequality. In addition, even if exacerbatory school mechanisms do outweigh compensatory ones, we still need to know if the inequality produced by schools is worse than that produced by non-school environments.

We call for a more contextual understanding of how schools matter, a view that was initially undermined by the No Child Left Behind legislation. When No Child Left Behind was first implemented, it was clear that schools were to be evaluated on the basis of children's test scores at one

point in time, and there would not be any adjustments based on varying non-school environments. This made no sense, of course, given that between-school achievement gaps are significantly established at the onset of kindergarten. But to their credit, many states abandoned evaluations of schools based on test scores at one point in time and have moved toward value-added assessments, in which statistical models more persuasively isolate schools' contribution to children's learning. This change represents a noteworthy acknowledgment by policymakers that context matters.

Can this victory for the "context" side be expanded? Two more shifts are necessary. First, the public needs to understand that socioeconomic achievement gaps form primarily before formal schooling and schools probably do more to reduce than increase them. If the public and policymakers knew this, they likely would acknowledge that achievement gaps are generated and maintained primarily by forces outside a school's purview, and they might be more inclined to address the broader social conditions generating those gaps. Second, the public needs to understand that, once compared fairly, schools serving low SES children provide learning environments roughly on par with schools serving high SES children. When we say this, we do not mean to suggest that schools serving disadvantaged children are perfect and would not benefit from improvement or that certain practices within schools are not advantageous for high SES children. What we mean is that when school effects are carefully isolated, there is surprisingly little evidence that schools serving high SES children do a better job promoting math and reading learning than do schools serving low SES children. Sociologists should be at the forefront of making this more contextual understanding of schools better known.

We recognize that the U.S. public is exceptional in the way they view public investment in schools as a legitimate action of the welfare state yet are skeptical of state involvement via other means. People in the United States are more reluctant than Canadians or Europeans to help families via non-school policies (e.g., increasing the minimum wage, greater public transportation, more generous family leave policies, broader access to health care, pro-worker legislation, and reducing mass incarceration) (Katz 2013; McCall 2013). This cultural position has consequences and too often directs policy efforts toward schools when the source of the problem is elsewhere.

Education scholars have been complicit in promoting school reform as the answer. Based on evidence that almost certainly fails to separate school from non-school effects, they have encouraged the notion that schools serving advantaged children promote more learning than schools serving disadvantaged children, and they have downplayed the fact that skill gaps are mostly formed prior to schooling and schools, as currently constituted, do more to reduce than increase these gaps. They have highlighted how better teachers promote much more learning than weaker teachers and then implied that improving the teachers black students are exposed to could readily close the black/white achievement gap (Gordon et al. 2006) even though they lacked evidence that teachers serving black children are poorer than those serving white children. Economist Eric Hanushek (1992:106) explains the focus on schools: "While family inputs to education are indeed extremely important, the differential impacts of schools and teachers receive more attention when viewed from a policy viewpoint. This reflects simply that the characteristics of schools are generally more easily manipulated than what goes on in the family."

But should current political conditions regarding what is considered the easiest policy reform shape scholars' willingness to identify the real causes of achievement gaps? The assumption that policy affecting families and neighborhoods is out of bounds and school policy is the only lever available has been a significant obstacle to understanding how schools really matter. This focus promotes the narrative that social conditions outside of schools are somehow natural and independent from policy decisions. They are not (Fischer et al. 1996). Earlier we mentioned that Canadian 4- to 5-year-olds outperform U.S. children on reading tests before schools have a chance to matter. It is noteworthy that Canadians are ahead by the largest margin at the bottom of the performance distribution—the United States has a longer tail of poor performers (Merry 2013). The Canadian people have made a wide range of different policy decisions that have made it easier to be poor in Canada than in the United States.

A cynical—and entirely plausible—interpretation of the greater emphasis on reforming schools relative to reforming other institutions is that it serves a purpose: It distracts people from the real sources of inequality, thereby serving the interests of those who benefit from current social

arrangements. Spring (2013:14), for instance, posits that school-based reform is inherently conservative because it avoids the more central conflict between the interests of labor and capitalists: "One reason schools have gotten involved with so many social problems is that the school is the most available institution and the one least likely to affect other parts of the social system." If this is so, then school policy provides an arena where reformers' energies can be "cooled out" with little chance of undermining existing power structures.

The current weakness in the sociology of education literature—underestimating the compensatory role of schools—stems from unwarranted certainty among scholars that schools promote inequality. It would be equally problematic to replace that position with one that contends that schools are uniformly compensatory, especially because currently we only have evidence that schools are compensatory with regard to socioeconomic gaps in cognitive skills and BMI and that they play a neutral role with respect to social and behavioral skills. Instead, we call for a more balanced understanding of the relationship between schools and inequality, one where neutral, exacerbatory, and compensatory possibilities are all given serious consideration and scholars acknowledge that inequality between groups is already on a strong trajectory when children arrive at kindergarten. We need to know more about how schools *refract* that trajectory. In the past half century, we have made less progress on this question than we should have. In the next 50 years, we can do better by rigorously isolating school effects, carefully weighing the magnitude of exacerbatory and compensatory mechanisms, and placing school effects in context.

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